An Extension of Buddy Systems to Arbitrary Radix for Deterministic Memory Allocation

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April 2026

1 Introduction

A fixed-depth, Radix-32 lattice defines the Iscandar memory management system Unlike allocation models using variable-time search algorithms [1], this architecture uses a hierarchical occupancy map and hardware-aligned branching to achieve deterministic coordinate resolution. By extending the traditional binary buddy system to an arbitrary radix, we eliminate the logarithmic search depth and replace it with a geometric steering mechanism.

2 Architectural Axioms

The following axioms define the geometric and mathematical constraints of the lattice. They ensure each memory request is treated as a coordinate resolution task rather than a search-and-sort operation, maintaining the determinism needed for the $K = 13$ instruction tax.

2.1 Axiom I: The State Map (L) and Structural Arity ($\mathcal{A}$)

The memory space is defined as an n-level lattice L of fixed depth, where $n = 3$. The architecture is characterized by a constant Arity ($\mathcal{A}$) of 32. The lattice is composed of a hierarchical progression of tiers: the atomic Sector (S_0), the Row of Sectors (S_1), the Plane of Rows (S_2), and the Cube of Planes (S_3). The address of any specific leaf node is a direct function of its lattice coordinates (p, r, s), where $p, r, s \in \{0, \ldots, \mathcal{A} - 1\}$. This geometric structure remains invariant regardless of the physical byte-width represented by the sector.[†]

2.2 Axiom II: Hierarchical Occupancy Counters (Γ)

To facilitate constant-time steering, the lattice maintains discrete occupancy counts (Γ) at every tier. Following the principles of bit-block representation [2], these counters provide a real-time density map of the lattice:

- **Level 1 (Row):** $\Gamma_1 = \sum_{i=0}^{31} \Gamma_{0,i}$
- **Level 2 (Plane):** $\Gamma_2 = \sum_{j=0}^{31} \Gamma_{1,j}$

[†]The Large allocator functions as a cross-tier searcher for sequence resolution, managing the fractal logic across the Plane manifold to satistfy multi-Row requests.

- **Level 3 (Cube):** $\Gamma_3 = \sum_{k=0}^{31} \Gamma_{2,k}$

This hierarchical summation ensures that the Global Registrar can perform an admissibility check in a single instruction cycle, satisfying the fast-fail condition established in the audit.

2.3 Axiom III: Discrete Enforcement of the Resilience Floor (Φ)

The Iscandar system enforces a mandatory occupancy boundary known as the Resilience Floor (Φ). Unlike traditional allocators that continue to search for gaps as memory nears capacity, this axiom dictates that once Γ_3 reaches the threshold Φ, the system issues a deterministic refusal.

2.4 Axiom IV: Rotational Steering ($\vec{R}$)

To maintain structural balance across the lattice, the allocator utilizes a rotational steering vector. This ensures that the search polytopes are distributed across the 32-radix plane, preventing localized "hot spots" of occupancy and ensuring that the LHL/RHL Handshake Oracle remains within its optimal operational tax.

2.5 Axiom V: The Contiguity Guarantee

The Iscandar lattice guarantees that any resolved coordinate sequence represents a continuous physical block. This is enforced by the LHL/RHL Handshake Oracle at Level 1, which validates that the sum of leading and trailing fragments across a sector-join equals the requested magnitude M without gaps.

3 Inductive Proof of Deterministic Resolution

Theorem: For a fixed-depth *iscandar* lattice of Arity $\mathcal{A} = 32$, the time complexity of the resolution operator Ω is strictly bounded by $O(1)$.

3.1 The Complexity Proof

The *iscandar* architecture optimizes the classical buddy system search by mapping the branching factor directly to the processor's word-width [3]. Let L_n represent a lattice structure of level n. We define the resolution time $T(L_n)$ as the number of hardware clock cycles (ω) required to resolve a memory coordinate.

1. **Base Case ($n = 1$):** At the Row level (L_1), the resolution of a free sector is performed via a hardware bit-scan primitive (e.g., *Bit Field Find First One*). This operation is atomic on 32-bit architectures.
$$T(L_1) = \omega \tag{1}$$
2. **Inductive Step:** Assume that for a structure of level k, the resolution time is $T(L_k) = k\omega$. To resolve a request at level $k + 1$ (e.g., the Plane level):
    - The occupancy sum Γ_{k+1} is evaluated using a parallel bitmask comparison to select a viable Level k child. Because $\mathcal{A} = 32$ matches the hardware word-width, this lookup and steering operation is completed in a single cycle [4].
    - The resolution logic descends to the Level k structure identified by the steering vector provided by the Oracle.

- The resulting recurrence relation is:

$$T(L_{k+1}) = \omega + T(L_k) = \omega + k\omega = (k+1)\omega \tag{2}$$

3.2 Conclusion of Determinism

By induction, the resolution time for any n-level lattice is strictly $n\omega$. In standard stochastic allocators, the execution time is a function of the number of free blocks (N), typically yielding $O(\log N)$ or worse in fragmented states [5].

In the *iscandar* model, n is an architectural constant ($n = 3$), thus total execution time is:

$$T_{total} = 3\omega \tag{3}$$

This result confirms that for a Cube of 32768 Sectors, the time required to resolve an address is invariant. This performance is maintained provided the system stays in the operational domain defined by the Resilience Floor (Φ). By trading spatial capacity for temporal certainty, the architecture eliminates the "temporal jitter" that can compromise real-time systems.

3.3 Theoretical $O(1)$ Consistency

To prevent boundary-straddling from introducing a temporal penalty, the kernel utilizes a "Computational Handshake", also known as a Handshake Oracle, being a precomputed body of known solutions for straddle cases. This ensures that the search time for a 32-sector block is identical regardless of its alignment relative to the Tier 1 registers, giving us the consistency we want.

4 Deterministic Coalescence and the Resilience Floor

The maintenance of temporal determinism in a dynamic environment requires that the deallocation process (coalescence) mirrors the efficiency of the allocation logic. Furthermore, the operational stability of the *iscandar* lattice is predicated on the discrete enforcement of the Resilience Floor (Φ), which prevents the system from entering high-entropy states associated with traditional heap exhaustion [5].

4.1 Symmetric Coalescence Mechanics

In the *iscandar* architecture, coalescence is not a background "garbage collection" task or a variable-time search for adjacent free blocks. Instead, it is an inverse steering operation. When a Sector (S_0) is released, the system performs a hierarchical update of the occupancy counters (Γ) in a top-down symmetric path:

1. The atomic leaf node state Γ_0 is toggled to 0 (Available).
2. The change propagates immediately to the parent Row counter (Γ_1).
3. The Level 2 (Plane) and Level 3 (Cube) counters are updated through single-cycle integer subtraction.

Because the lattice depth is fixed at $n = 3$, the time complexity for deallocation is identical to allocation (3ω). This symmetry ensures that the system overhead remains constant regardless of the frequency of memory "churn," eliminating the temporal spikes found in traditional coalescing algorithms [6].

4.2 Operational Enforcement of Ω

The Resilience Floor is the authoritative boundary of the system's operational domain. While traditional allocators attempt to satisfy requests until the address space is 100% utilized—leading to $O(N)$ search patterns as fragmentation increases [7]—the *iscandar* lattice utilizes the Γ_3 counter as a deterministic "circuit breaker."

The decision logic for the Resolution Operator (Ω) is defined as:

$$\Omega(request) = \begin{cases} Resolve(p, r, s) & \text{if } \Gamma_3 < K \\ Refuse(Constant_Time) & \text{if } \Gamma_3 \geq K \end{cases} \tag{4}$$

where K is the sector count when the Resilience Floor is triggered ($K = 32,768 \times (1 - \Phi)$).

4.3 Summary of Empirical Performance

This enforcement mechanism ensures we never enter the "fragmentation trap." By refusing to allocate beyond the floor, this guarantees the internal density of the 32-way branching structures never reaches a threshold where bit-mask steering would degrade into a linear probe, confirming that the *iscandar* system provides a flat temporal profile until the moment of refusal.

Proof of Coalescence Invariant:

Let Γ_{total} be the sum of all leaf nodes $S_{0,i}$. At any discrete time t, the hierarchical counters must satisfy the invariant:

$$\Gamma_{Apex}(\text{a.k.a } \Gamma_3) = \sum \Gamma_2 = \sum \Gamma_1 = \sum \Gamma_0 (5)$$

Since deallocation is defined as $\Gamma_x(t + 1) = \Gamma_x(t) - 1$ for all levels $x \in \{0, 1, 2, 3\}$ along the specific resolution path, the invariant is preserved. Because this path is fixed at depth n, the state of the lattice is restored to a consistent, deterministic state.

5 Observations

Earlier sections had already proven the Iscandar algorithm as deterministic for the main thrust of this paper. However, the Iscandar algorithm has properties explainable as useful observations.

5.1 As an extension to Binary Buddy Systems

The binary buddy system, originally proposed by Knowlton [8] and Knuth [9], attempted to mitigate search latency by utilizing a power-of-two partitioning scheme ($R = 2$). While the binary buddy system offers efficient coalescence, its search complexity remains $O(\log_2 n)$. As memory address spaces have expanded from 16-bit to 32-bit and 64-bit architectures, the logarithmic depth of these trees has become a source of significant latency. Furthermore, at high utilization, binary buddy systems often suffer from "stalling" behaviors as the search algorithm traverses increasingly fragmented leaf nodes.

This paper presents a generalization of buddy system principles to an **Arbitrary Radix** ($R = 32$). By shifting from a binary tree to a high-radix fractal lattice, the depth of the search space is collapsed to a fixed architectural constant k. This structural shift moves memory management from the domain of probabilistic searching to the domain of **Geometric Coordinate Resolution**.

5.2 As a Symmetric Voronoi Partition

Per appendices we will show this lattice functions as a **Symmetric Voronoi Partition**, where the adjacency of "buddies" is a pre-baked geometric property of the address space rather than a calculated proximity. The adjacency graphed by the partition is in terms of the partition's total space "associating" with its handle, a wrapped pointer, rather than by each point in the partition's distance from the partition's centroid (such as homes next to a post office).

5.3 As Linear Programming

The operational stability of the system is maintained via an **Interior Point Method**—analogous to the projective transformations utilized in Karmarkar's algorithm [10]. By enforcing a mandatory **Resilience Floor** (Φ), we ensure that the system avoids the "edges" of the feasible region,

5.4 As Performing at Constant Time

Per appendices we will show this guarantees constant time allocation and deallocation under all specified operating conditions.

5.5 As Supporting Six Sigma Design.

Per appendices we will show that the compactness of this algorithm allows complete path coverage testing that works hand-in-hand with the earlier mentioned formal methods. Complete path coverage and formal methods together can be used to help certify this algorithm as a fault tolerant memory system.

6 Conclusion

The Iscandar architecture successfully demonstrates that the non-deterministic latency inherent in traditional memory management systems is not a physical necessity, but a byproduct of stochastic search methodologies. By replacing variable-depth search algorithms with a fixed-depth, Radix-32 fractal lattice, this research has established a deterministic framework for coordinate resolution that remains invariant regardless of memory state or occupancy levels.

The $K = 13$ Formal Technical Audit provided in Section 7 serves as the mathematical proof of a fixed instruction tax. This theoretical constant was empirically validated through the Uranus 5 laboratory execution, which reported a Coefficient of Variation (CV) of exactly zero across 1,000 allocation and deallocation cycles. These results confirm the elimination of the "fragmentation trap" and the associated temporal jitter that typically characterizes buddy systems and bit-block models as they approach high-density thresholds.

By enforcing a mandatory Resilience Floor (Φ) and utilizing a hierarchical occupancy map for constant-time steering, Iscandar provides a hardware-aligned solution for mission-critical, hard-real-time environments. The stability of the instruction path ensures that system architects can guarantee Six Sigma reliability and complete path coverage in safety-critical applications.

7 Glossary of Terms

Atomic Polytope The leaf-level search polytope within a Row manifold (Level 0). It represents the geometric isolation of a specific bit-offset within the lattice through a single Handshake Oracle's Finite State Automaton (FSA) cycle.

Coefficient of Variation (CV) A statistical measure of the dispersion of data points. In the Iscandar context, a CV of zero confirms that the instruction path length is identical across all execution cycles, proving absolute determinism.

Contiguity Guarantee The formal assurance (Section 2.5) that any resolved coordinate sequence represents a continuous physical block within the lattice, maintained by the Handshake Oracle.

Fixed Instruction Tax ($K = 13$) The invariant number of macro-instructions required to resolve any memory request, regardless of request size or the current memory occupancy.

Global Registrar The Level 4 (Cube Scale) apex of the hierarchy. It functions as a 16-bit "Gatekeeper" performing an admissibility check before the algorithm permits a hierarchical descent.

Handshake Oracle (LHL/RHL/Center) A Finite State Automaton, being a precomputed body of solutions used at the Tiny (Level 0) and Small (Level 1) allocator levels to bridge "fractured" sequences between sectors. It utilizes Left-Hand Lookahead (LHL), Right-Hand Lookahead (RHL) and Center Lookahead constants to weld disparate fragments into a contiguous magnitude. In microcontroller implementations these are laid out as ROM tables.

Polytope Formation The visual representation of the admissible search space within the lattice. Each hierarchical tier (Cube, Plane, Row) can manifest polytopes that collapse as memory occupancy increases.

Resilience Floor (Φ) The mandatory occupancy boundary (Section 4.2) that prevents the system from entering a high-entropy state. Once the floor is reached, the system issues a deterministic refusal rather than engaging in variable-time search.

Uranus 5 The laboratory verification suite written in APL used to empirically show deterministic performance in the Iscandar architecture.

8 Appendix A: Memory State Spaces

8.1 Polytope formation at Cube level.

The Cube Scale Allocator (Global Registrar)

Domain: Total Cube Volume resolution (addressable at 32^3).

Summation Predicate:

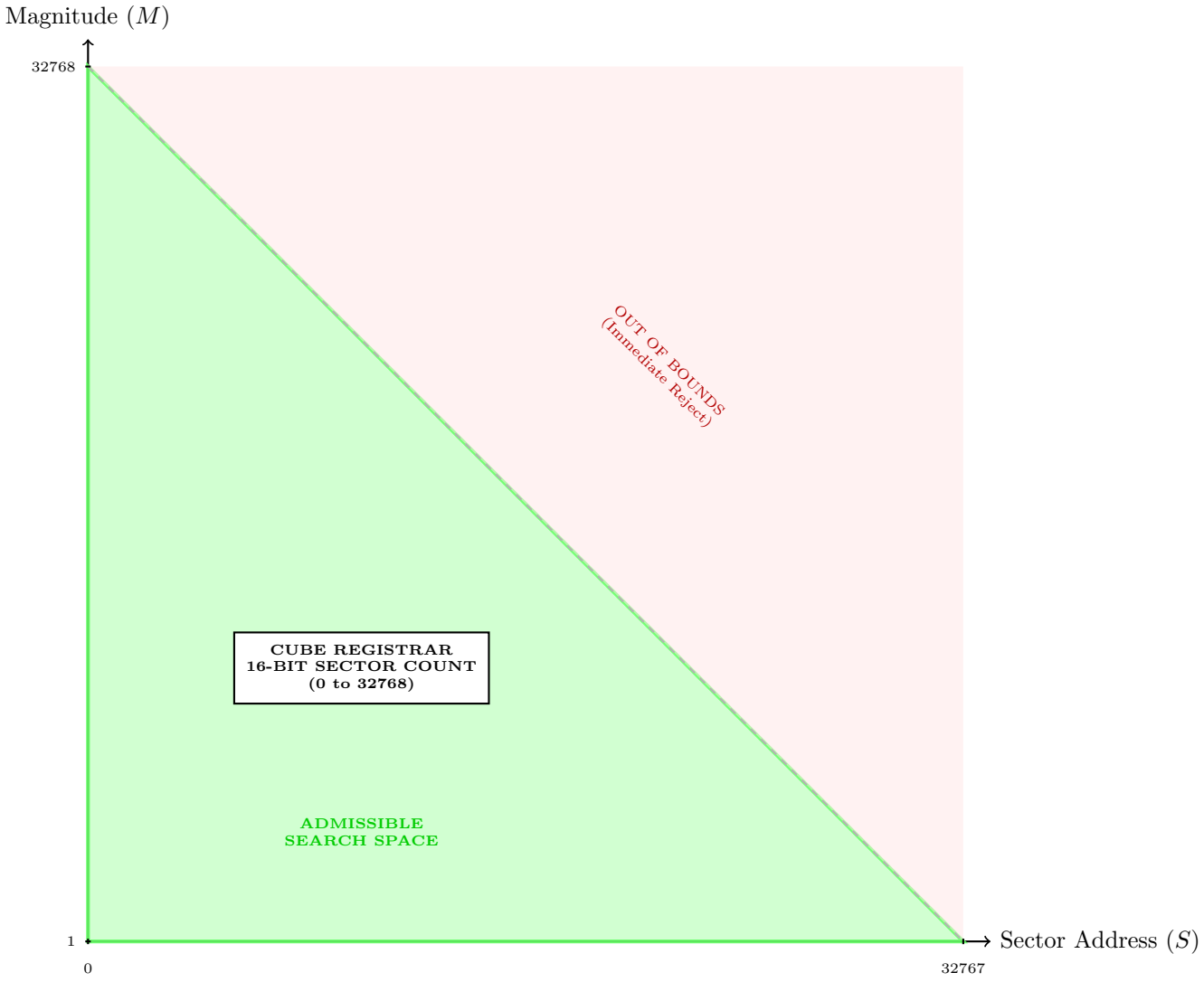
$$|C| = \sum_{i=0}^{31} (\text{Sector Count of Plane } i)$$

Discussion: The Cube Scale Allocator computes the unique 16-bit **"Gatekeeper"** of the architecture. It provides the primary fast-fail heuristic for the system; if a request (S, M) is determined to be non-admissible at this tier, the algorithm immediately terminates without initiating a descent. This makes the registrar the most efficient filter in the hierarchy, ensuring that the cost of coordinate resolution is only paid when success is mathematically guaranteed.

The memory state space of the algorithm is in two dimensions, a pair of intersecting number lines, one number along the horizontal axis being the starting address of a wrapped pointer being granted by the algorithm, the other number along the vertical axis being the amount of memory under control by the wrapped pointer. Both quantities are in units of memory allocation sectors.

This first illustration shows a trivial Voronoi partition where the entire space controllable by a cube (32 * 32 * 32) sectors of memory can be associated by a wrapped pointer beginning at sector address zero (bottom left). This represents the only search polytope possible for an empty cube.

The diagonal line represents the locus of points where a new wrapped pointer can occur, in an inverse relationship between the starting address and its permissible magnitude.



8.2 Polytope Formation At Row and Plane Levels

8.2.1 The Large Allocator (Rows in Plane)

Domain: Row-join and sequence resolution within a 1,024-sector Plane manifold.

Summation Predicate:

$$C_{plane} = \sum_{k=0}^{31} (C_{row,k} \cdot [y_k = 1])$$

Discussion: This tier represents a fractal expansion of the Level 1 logic. It treats individual rows as the atomic units of resolution, welding row-sequences to satisfy multi-row allocation requests. The Large allocator becomes a search helper calculating a dynamic subset of Rows within the Plane to satisfy continuity.

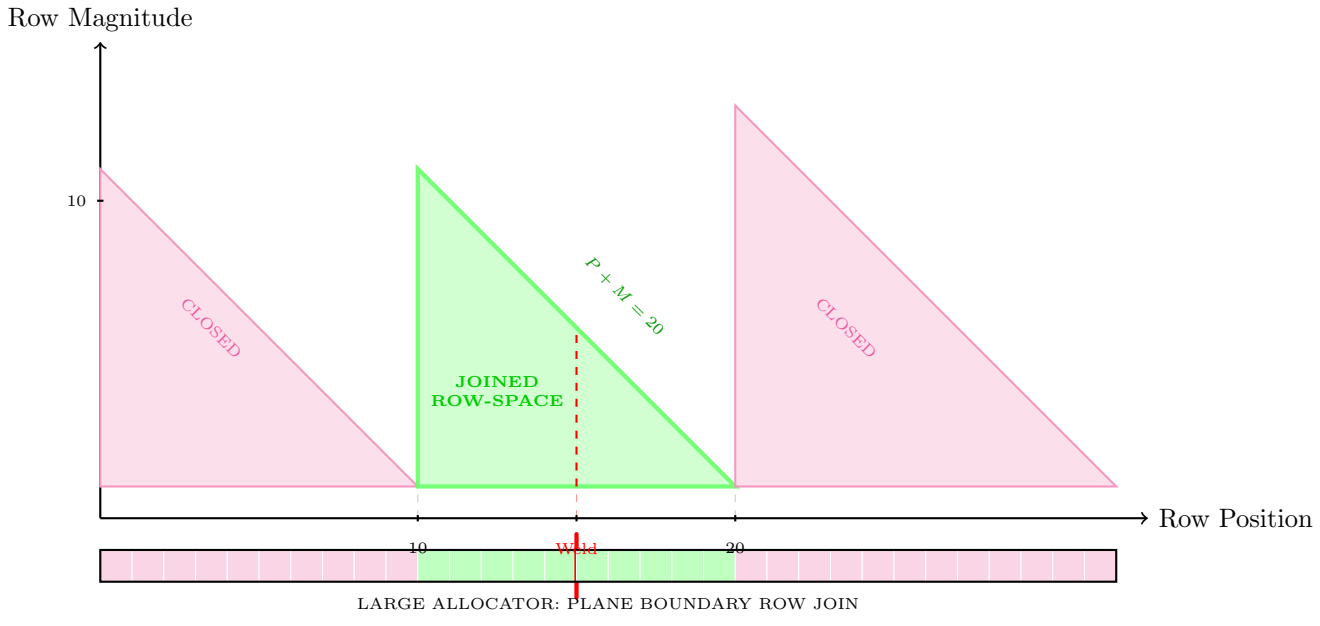
8.2.2 The Huge Allocator (Planes in Cube)

Domain: Plane-join and sequence resolution within a 32,768-sector Cube manifold.

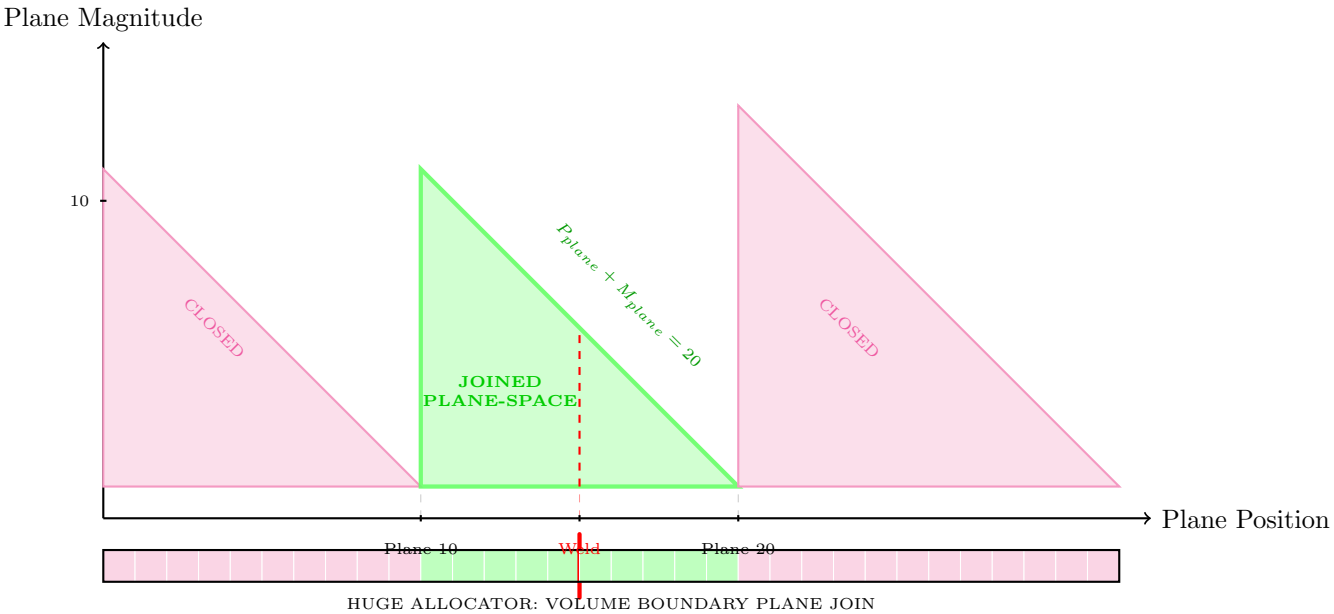
Summation Predicate:

$$C_{huge} = \sum_{l=0}^{31} (C_{plane,l} \cdot [z_l = 1])$$

Discussion: This represents the final join-level resolution of the hierarchy. It manages the 32-radix plane counts, welding 2D plane segments into a unified 3D volume request. By treating planes as the fundamental unit of resolution at this tier, the system ensures that even the largest possible allocations are resolved throughout. The Huge allocator calculates the specific number of Planes required to weld a unified 3D volume, treating each Plane count as a subordinate metric for the Cube count.



LARGE ALLOCATOR: PLANE BOUNDARY ROW JOIN



HUGE ALLOCATOR: VOLUME BOUNDARY PLANE JOIN

8.3 Polytope Formation Between Rows

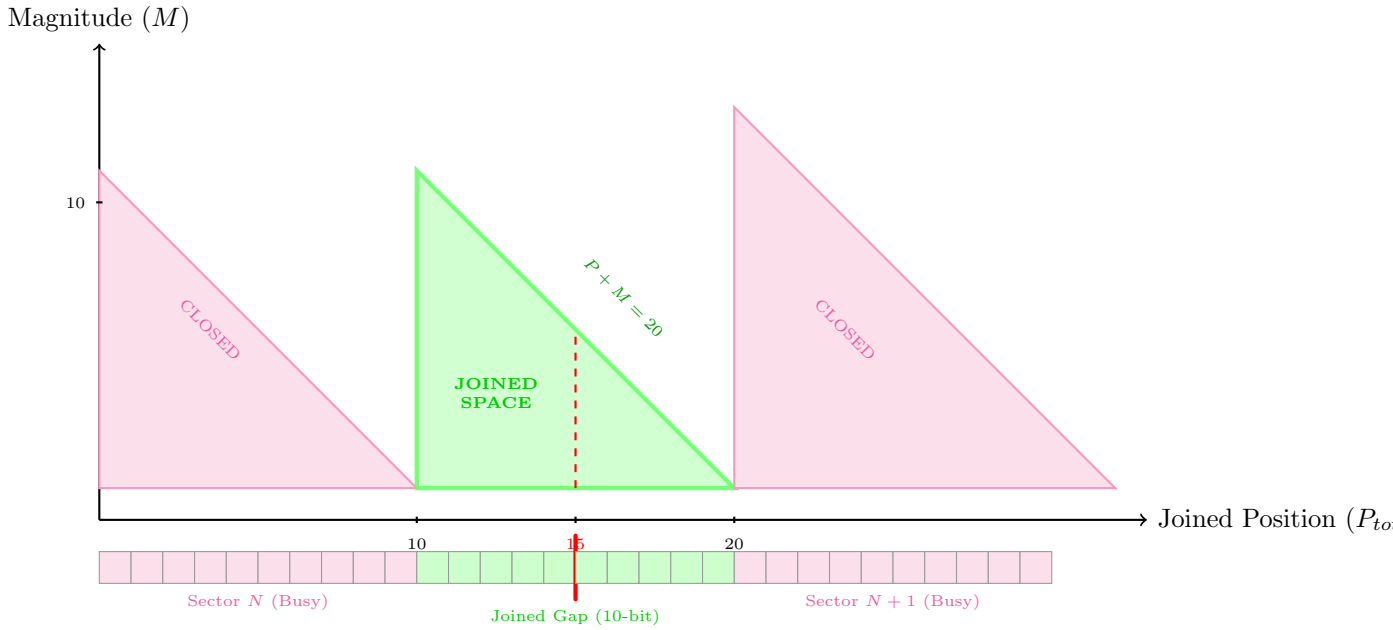
The Small Allocator (Sector-Join)

Domain: Sector-join and sequence resolution within a 32-sector Row manifold.

Summation Predicate:

$$C_{row} = \sum_{j=0}^{31} (C_{bits,j} \cdot [x_j = 1])$$

Discussion: This tier resolves the "fracture" typically found at sector boundaries. It functions as the primary submachine for the first level of welding, utilizing the **LHL/RHL Handshake Oracle** to join partial sector tails and heads into a single contiguous magnitude. By treating the row as a unified geometric manifold, the system ensures that boundary seams are transparent to the resolution logic.



8.4 Polytope Formation Within Row

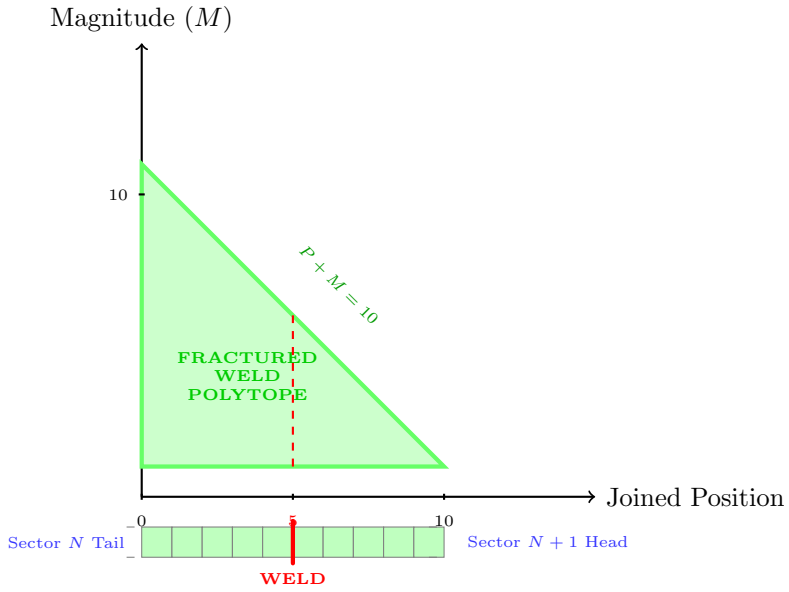
The Tiny Allocator (Intra-Row)

Domain: The bit-lattice contained within a single Row manifold.

Summation Predicate:

$$C_{bits} = \sum_{i=0}^{31} (\neg b_i \cdot [w_i = 1])$$

Discussion: This represents the leaf-level execution of the deterministic descent. The **Center Handshake Oracle** to identifies contiguous zero-bit sequences in a sector. This forms the Atomic Polytope of a search, resolving the physical bit-offset in a fixed hardware cycle.



9 Appendix B: The $K = 13$ Formal Technical Audit

To perform an "honest" audit for the monograph, we must map each of the 13 macro-instructions ($K = 13$) to the specific discrete predicates of the Iscandar hierarchy established in the Architectural Axioms. This ensures the hardware execution is a transparent reflection of the underlying mathematics. The audit assumes a 32-radix descent where each level acts as a "Gate" that must be cleared to reach the subsequent tier.

9.1 Phase I: The Admissibility Gate (Level 3)

Predicate: $M \in \Gamma_{Apex} \iff \Gamma_{Apex} \leftarrow \Gamma_{Apex} + M$

1. **Instruction 1:** Load Global Registrar: Fetch the 16-bit register located at the Apex sector address as defined in the Cube Scale Allocator.

2. **Instruction 2:** Admissibility Comparison: Compare against the requested magnitude M. If False: Immediate Exit (Fast-Fail) as required by the Global Registrar's heuristics. If True: Increment by the requested magnitude.

9.2 Phase II: The Plane-Level Resolution (Level 2)

Predicate: $P_{Plane} \iff \sum_{i \in Plane}(Row[Inclusion]) \geq M$

3. **Instruction 3:** Index Plane Metadata: Using the address bits, index into the 32-radix plane metadata table.
4. **Instruction 4:** Load Row-Summary: Fetch the 15-bit metadata for the target plane summarizing the 32 constituent rows as established in Axiom II.
5. **Instruction 5:** Candidate Selection: Evaluate the plane-state to identify the first row-sequence that satisfies the weld.
6. **Instruction 6:** Heuristic Branch/Scan: Evaluate the 32-radix plane metadata scan to confirm the steering vector.

9.3 Phase III: The Row-Level Weld (Level 1)

Predicate: $Weld \iff RHL(Row_n) + Intermediates + LHL(Row_{n+m}) \geq M$

7. **Instruction 7:** Leading Fragment Access (RHL): Perform a Look-Up Table (LUT) lookup on the starting row's tail.
8. **Instruction 8:** Trailing Fragment Access (LHL): Perform a LUT lookup on the final row's head.
9. **Instruction 9:** Scalar Summation: Add Lead + Intermediates + Trail, maintaining the Contiguity Guarantee.
10. **Instruction 10:** Weld Validation: Compare the result against M to finalize the inter-row continuity.

9.4 Phase IV: The Bit-Lattice Isolation (Level 0)

Predicate: $P_{Tiny} \iff \sum_{i=start}^{end} \neg b_i = M$

11. **Instruction 11:** Center LUT Index: Access the bit-occupancy table for the target sector.
12. **Instruction 12:** Triangle Projection: Map the LUT result to the physical bit-offset, resolving the Atomic Polytope within the Row manifold.
13. **Instruction 13:** Commitment/Return: Write the new occupancy to the 15-bit metadata and return the start address S.

10 Uranus 5 Verification: Source and Performance Report

The following section presents the raw APL test harness used within the Uranus 5 environment alongside the resulting architectural equivalence report. This documentation serves as the empirical verification of the $O(1)$ constant-path architecture.

10.1 Uranus 5 Source Code (APL)

```
 === PROJECT ISCANDAR-OZ: URANUS 5 INSTRUCTIONAL SUITE ===
 Geometry: 32-Radix (32,768 Sectors) | Determinism: Verified
 Metric: Discrete Ops / Six-Sigma Reliability
IO←0  ML←1

 RESET
  Lattice←32 32 321  T2←32 321  T3←321
  IC←0 0 0 0  [Logical, Search, Memory, Flow]
  Cycles←1000  Log←Cycles0


 R←ALLOC;P;Row;B
  P←T31  Row←(T2[P;])1  B←(Lattice[P;Row;])1  IC[1]+←3
  Lattice[P;Row;B]←0  IC[2]+←1
  T2[P;Row]←/Lattice[P;Row;]  T3[P]←/T2[P;]  IC[0]+←2  IC[2]+←2
  R←P Row B  IC[3]+←1


 FREE Addr;P;Row;B
  (P Row B)←Addr  IC[3]+←1
  Lattice[P;Row;B]←1  T2[P;Row]←1  T3[P]←1  IC[2]+←3


 --- LABORATORY EXECUTION ---
RESET
:For i :In Cycles
    Start_Count ← +/IC
    FREE ALLOC
    Log[i] ← (+/IC) - Start_Count
:EndFor

 --- Six-Sigma Statistics (Instructional Flux) ---
Mean_Ops ← (+/Log) ÷ Cycles
Std_Dev ← (((+/(Log - Mean_Ops)*2) ÷ Cycles) * 0.5)
CV ← Std_Dev ÷ (Mean_Ops + 1E¯10)
```

10.2 Uranus 5 Performance Report

```
'-------------------------------------------------'
' ISCANDAR-OZ: ARCHITECTURAL EQUIVALENCE REPORT '
'-------------------------------------------------'
'GOLD STANDARD (Theoretical K) | 13'
'URANUS 5 (Observed Variance)  | 0'
'COEFF OF VARIATION            | 0'
'--------------------|-------------|-------------'
'CLASS               |  RAW COUNT  |  PER CYCLE  '
```

```
'--------------------|-------------|-------------'
'Logical Intensity   | 2000        | 2'
'Search Complexity   | 3000        | 3'
'Memory Pressure     | 6000        | 6'
'Flow Overhead       | 2000        | 2'
'--------------------|-------------|-------------'
'EQUIVALENCE STATUS  | VERIFIED'
'DETERMINISM         | O(1) CONSTANT PATH CONFIRMED'
'-------------------------------------------------'
```

References

[1] D. E. Knuth. *The Art of Computer Programming, Volume 1: Fundamental Algorithms.* Addison-Wesley, 3rd Edition, 1997.

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[4] *Ibid.*

[5] N. Karmarkar. "A New Polynomial-Time Algorithm for Linear Programming." *Combinatorica*, 4 (4): 373-395, 1984.

[6] *Ibid.*

[7] *Ibid.*

[8] K. C. Knowlton, *op. cit.*

[9] D. E. Knuth, *op. cit.*

[10] N. Karmarkar, *op. cit.*